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Models and Tools for Cyber-Physical Systems
Exercise sheet 1
WITH SOLUTIONS

Exercise 1: UPPAAL introduction

In this exercise, we show how to encode the Switch extended state machine (Chapter 2, Slide 25) in the tool UPPAAL. Then we make some modifications and analyze the model.

(a) Download UPPAAL from `https://uppaal.org/` in version 5.0.0. Also download the UPPAAL Switch model from Moodle. Then open the model in UPPAAL.

(b) Watch the introductory video to UPPAAL, which can be found in Moodle.

(c) Modify the UPPAAL model such that the process `User` turns the light on and off (i.e., sets `press = 1`) every three rounds.

(d) In your modified version, what is the maximal reachable value for local variable `x`?

(e) Verify that your model is deadlock-free.

Hint: Use the predefined queries in the *Verifier* tab for tasks (d) and (e).

.....Solution sketch .....

Download the UPPAAL model solution from Moodle.

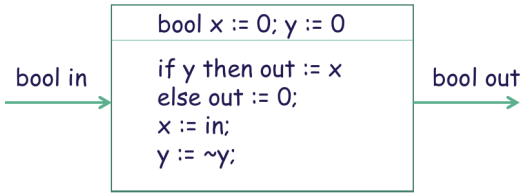
(c) The new component `User3` achieves the goal by adding a counter and switching when the counter reaches the value 3. (Note that we had to replace the old `User` component in the system declarations.)

(d) We can use the first two verifier queries to check that the value 2 can be reached but not exceeded.

(e) The third query verifies deadlock freedom.

Exercise 2: Extended state machine

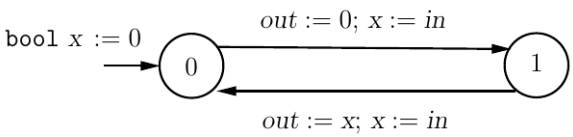
Consider the component `OddDelay` (Chapter 2, Slide 24):



Draw an extended state machine for this component with two modes and one state variable for `x` (you may also call it `x`). Use the mode of the state machine to capture the value of the component's state variable `y`, while the state variable `x` should be updated using assignments in the mode switches.

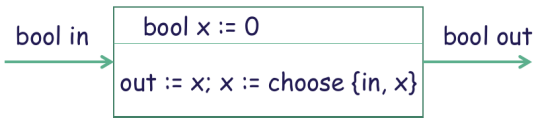
.....Solution sketch .....

The extended state machine corresponding to the component `OddDelay` is shown below. The modes correspond to the values of the state variable `y`.



Exercise 3: Mealy machine

Consider the nondeterministic version of the `Delay` component (Chapter 2, Slide 18):

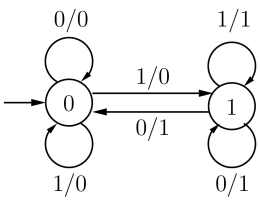


Draw the Mealy machine corresponding to the component and describe how the nondeterminism manifests in the Mealy machine.

(Remember that each state of the Mealy machine corresponds to a state of the original component, i.e., a valuation of the state variable `x`, and each transition contains an input and the corresponding output.)

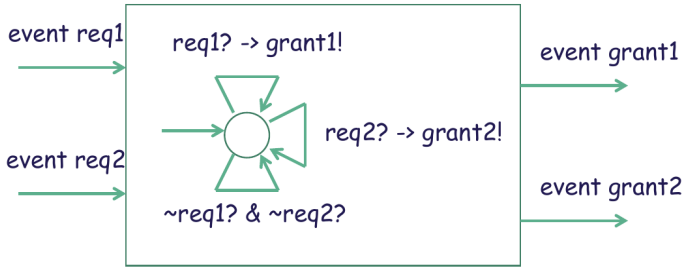
.....Solution sketch .....

There is only one state variable (`x`) of type `bool`, i.e., there are two states, which we will call `0` and `1`. The corresponding two-state Mealy machine is shown below. Note that the component is nondeterministic whenever the value of the input (`in`) differs from the value of the state variable `x`. We can see the nondeterminism when there are two possible transitions for input 0 in mode `1` (and similarly for input 1 in mode `0`).



Exercise 4: Reaction description

Consider the nondeterministic component Arbiter (Chapter 2, Slide 35):



The reactions are expressed using the notation of extended state machines. Write an equivalent description using pseudocode. You can use a local variable whose value is assigned nondeterministically with the `choose` construct.

.....Solution sketch .....

The following code can be used as the reaction description of the component `Arbiter`. When both input request events are present, the value of the local variable `b` is chosen nondeterministically and is then used to decide whether to issue the output event `grant1` or `grant2`.

```
if req1? then
   if req2? then
      local bool b := choose(0, 1)
      if b then
        grant1!
      else
        grant2!
   else
     grant1!
else
  if req2? then
    grant2!
```